

Baohua Yang

Principal Architect | Enterprise AI, Agentic Systems & Distributed Platforms
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PROFILE

- Principal Architect and technical leader with 15+ years of research and engineering experience in distributed systems and enterprise platforms, with a current focus on enterprise AI and agentic systems.
- Designs and delivers enterprise AI and agentic systems, combining product strategy, system architecture, and hands-on engineering.
- Key contributor and maintainer across Hyperledger, OpenStack, and Docker; active mentor in the open-source community.

TECHNICAL SKILLS

- AI: multi-agent orchestration (LangChain/LangGraph), RAG, MCP, context management, evaluation & telemetry, data protection, LLM inference, fine-tuning, guardrails.
- Languages & Systems: Python, Go, C/C++, Kubernetes, distributed infrastructure.

EXPERIENCE

Principal Architect, *Oracle*

Aug 2017 – Present

- Led and built enterprise AI and distributed-system products for Fortune 500 customers, spanning a ~\$200M ARR portfolio and providing technical leadership across ~50 engineers.
- Designed and implemented an enterprise AI agent, improving query-result accuracy from ~30% to 90+% and cutting response latency by 50%.
- Influenced the architecture and feature roadmap of Oracle Select AI Agent, helping deliver RAG-based querying, context management, and multi-agent orchestration.
- Designed multi-tenant isolation, resource quotas and autoscaling for Oracle Blockchain Platform on containerized Linux infrastructure, across Kubernetes clusters running several thousand pods.
- Set technical direction across AI, security, data, and platform architecture; partnered with other product and go-to-market teams to bring research into production.

Research Scientist, *IBM*

Jan 2013 – Aug 2017

- Architected and built enterprise blockchain platforms in Go, sustaining over 1,000 TPS per cluster in production, serving hundreds of distributed clusters.
- Core developer on Hyperledger Fabric; drove the applied research that fed the product.

OPEN SOURCE & COMMUNITY

Hyperledger Foundation TSC Member (now LF Decentralized Trust) 2017-2019; 2020-2021

- Helped shape technical strategy and project governance across the Hyperledger ecosystem; contributed to blockchain and distributed ledger standardization through ISO/TC 307.
- Mentored open-source developers and new contributors.

EDUCATION

Ph.D. in Control Science and Engineering, Tsinghua University Sep 2007 – Jan 2013

Research focus: high-performance distributed systems and algorithms.

Built a 10 Gbps unified threat management (UTM) system.

1st-class scholarship (2009) and IBM Ph.D. Fellowship Award (9 winners, 2011).

Visiting Scholar, EECS, University of California, Berkeley Sep 2010 – Sep 2011

Advanced distributed systems, including software-defined and data-center networking.

Bachelor of Science and Engineering, Tsinghua University Aug 2003 – Jul 2007

1st-class scholarship, 2006.

SELECTED PATENTS

- Methods and System to Support Long Context Management in AI Agentic System, 2026
- Enhanced NL2SQL AI Query System and Method, 2025
- Methods and System to Protect the AI Knowledge Base System, 2024
- [Systems and methods of transaction identification generation for transaction-based environment](#), US 11,822,538 B2 (granted 2023).
- [Methods and system of tracking transactions for distributed ledger](#), US 11,741,075 B2 (granted 2023).
- [Systems and methods of providing ledger as a service](#), US 11,546,425 B2 (granted 2023).
- [Self-adaptive building container images](#), US 10,983,816 B2 (granted 2021).
- 6 additional granted patents in distributed ledgers, networking, and mobile computing (10 granted patents total).

SELECTED PUBLICATIONS

- LazyCtrl: A Scalable Hybrid Network Control Plane Design for Cloud Data Centers. K. Zheng, L. Wang, B. Yang, Y. Sun, S. Uhlig. IEEE Trans. Parallel and Distributed Systems, 28(1):115–127, 2017.
- Keep Forwarding: Towards K-link Failure Resilient Routing. B. Yang, J. Liu, S. Shenker, J. Li, K. Zheng. IEEE INFOCOM 2014.
- Practical Multi-tuple Packet Classification using Dynamic Discrete Bit Selection. B. Yang, J. Fong, W. Jiang, Y. Xue, J. Li. IEEE Transactions on Computers, 63(2):424–434, 2014.
- Data-Driven Network Connectivity. J. Liu, B. Yang, S. Shenker, M. Schapira. ACM HotNets-X, 2011.

SELECTED BOOKS AND TECHNICAL GUIDES

- Harness Engineering Guide and Agentic AI Guide (2026)
- Docker Container Technology in Action, China Machine Press, 2015 (4 editions).
- Blockchain: Principle, Design and Applications, China Machine Press, 2017 (2 editions).